Stroma World Spec — §6: Assumptions & Exclusions

> **What this section is:** A complete, honest inventory of everything v1.0 gets wrong on purpose. **Why it matters:** Every simulation is a lie with a budget. This section discloses the lies, explains why they don't invalidate the stroma barrier question, and flags the ones most likely to bite us first.
> **Organization:** Assumptions (things we model in a simplified way) followed by Exclusions (things we don't model at all). Each entry states what's simplified or missing, why it's acceptable for the stroma barrier problem, what risk it carries, and when it should be revisited.

Part 1: Assumptions (Simplified Biology)

These are aspects of PDAC biology that exist in the simulation but in a form we know is incomplete. The simplification is deliberate and bounded.

A1. All tumor cells share one genotype.

What we assume: Every tumor cell in a run carries the same 21-gene state array. There is no intra-tumor heterogeneity, no subclones, no mutation during the simulation.

What's really true: PDAC tumors are genomically heterogeneous. Subclonal populations with different driver mutations coexist. Clonal evolution under therapy selects for resistant subclones.

Why this is acceptable: The stroma barrier problem is primarily a *microenvironment* problem, not a *genomic diversity* problem. The barrier forms because of what the tumor secretes (driven by KRAS, which is clonal in >95% of PDAC), not because of subclonal genetic variation. A monoclonal tumor will build the same barrier as a polyclonal one because they share the same KRAS-driven paracrine output. The EA is optimizing interventions against the barrier, not against clonal evolution — different problem, different cycle.

Risk if wrong: If resistance to the EA's discovered interventions depends on pre-existing rare subclones (rather than on inducible phenotypic adaptation via NRF2/ABCB1, which we do model), then v1.0 will overestimate intervention efficacy. The EA might find strategies that work against a uniform population but fail against a heterogeneous one.

When to revisit: Cycle 3 (Clonal Evolution). The architecture already supports per-cell genotypes — `gene_state[21]` is tracked per agent, not per population. Adding stochastic mutation at division (Rule 12, currently deferred) is a parameter change, not a structural rewrite.

A2. CAF activation is irreversible.

What we assume: Once a PSC flips ACTA2=ON, it stays ON forever. There is no CAF → PSC reversion, no CAF subtype switching (myCAF ↔ iCAF ↔ apCAF), and no signal-dependent deactivation.

What's really true: CAFs exhibit plasticity. Different subtypes (myofibroblastic, inflammatory, antigen-presenting) have been identified in PDAC. Some evidence suggests CAFs can partially revert when activation signals are withdrawn, though the extent of reversibility in established PDAC stroma is debated.

Why this is acceptable: The stroma barrier is built and maintained primarily by myofibroblastic CAFs (the ECM-producing phenotype). Our single CAF type approximates this dominant subtype. Irreversible activation ensures the barrier is durable — which matches clinical observation. PDAC stroma persists even after neoadjuvant chemotherapy that substantially reduces tumor cellularity. CAF plasticity matters most for immune modulation (iCAFs secrete different cytokines than myCAFs), which is a Cycle 2 problem. For barrier formation and drug exclusion alone, the irreversible myCAF approximation is faithful enough.

Risk if wrong: If the EA discovers an intervention that would, in reality, cause CAF deactivation (removing all TGF-β and SHH), the simulation will overestimate barrier persistence because CAFs never revert. The EA may fail to discover "starve the CAFs" strategies that could work in vivo. This is a conservative error — the simulation is harder to beat than reality, not easier.

When to revisit: Cycle 2 (CAF Subtypes). Add `activation_mode` enum with {PSC, myCAF, iCAF} and signal-dependent transitions. The module architecture (Module 3) already has an activation state variable; expanding its state space is straightforward.

A3. ECM is a single density field with two component fractions, not a fiber-resolved structure.

What we assume: ECM at each voxel is described by one scalar density (0–1) plus an HA/collagen fraction split. There is no fiber orientation, no crosslinking state, no pore size distribution, no viscoelasticity.

What's really true: Real PDAC stroma has aligned collagen fibers (TACS-3 signatures), heterogeneous pore geometry, variable crosslinking (LOX-mediated), and complex mechanical properties that depend on strain rate and direction.

Why this is acceptable: The stroma barrier's primary function in the simulation is to impede diffusion and provide mechanical confinement. Both effects depend on ECM density as the dominant variable. Fiber orientation matters for directed cell migration (collagen "highways"), but v1.0's motility model is not directionally biased — cells move faster or slower based on EMT state, not along fiber tracks. The HA/collagen split captures the two major physical effects (diffusion impedance vs. structural rigidity) without the computational cost of a full fiber-resolved model. For the question "can drug get through the barrier," density is the right level of abstraction.

Risk if wrong: If the EA discovers strategies that depend on anisotropic drug delivery (e.g., exploiting aligned collagen channels to route drug along specific paths), the simulation will miss this because ECM is isotropic. The EA will also miss invasion directionality along collagen fibers — but invasion dynamics are secondary to the barrier diffusion problem.

When to revisit: When invasion directionality or immune cell contact guidance becomes important (Cycle 2 for immune infiltration along collagen fibers, Cycle 3 for metastatic invasion). Consider adding a 2D orientation tensor per voxel at that point.

A4. Drug pharmacokinetics are single-compartment and generic.

What we assume: "Drug" is a single abstract substrate with one diffusion rate, one uptake rate, one kill curve, and one efflux response. It does not represent any specific chemotherapeutic agent. There is no metabolism, no prodrug activation, no multi-drug interaction, no pharmacokinetic modeling beyond simple diffusion and first-order uptake/decay.

What's really true: Gemcitabine (the PDAC standard) is a prodrug requiring intracellular phosphorylation. Nab-paclitaxel has entirely different delivery and PK characteristics. FOLFIRINOX is a four-drug combination with complex interactions. Real PK involves plasma clearance, tissue distribution, hepatic metabolism, and renal excretion.

Why this is acceptable: The stroma barrier problem is about *access*, not *pharmacology*. The question is: can the drug get through the ECM to reach tumor cells at sufficient concentration? The answer depends on ECM density and diffusion physics, not on whether the drug is gemcitabine or paclitaxel. A generic drug with tunable diffusion rate, kill threshold, and efflux susceptibility captures the essential competition between delivery and resistance. The EA's discoveries will be about *how to breach the barrier and maintain intracellular drug concentration*, not about which specific molecule to use.

Risk if wrong: Drug-specific effects matter for clinical translation. If the EA discovers "pulse drug for 6 hours, wait 18 hours," the optimal timing will be different for gemcitabine (short half-life) vs. nab-paclitaxel (long half-life). v1.0 findings will need to be re-validated with agent-specific PK models before clinical claims can be made. This is expected — v1.0 discovers strategic principles; later cycles validate them with realistic pharmacology.

When to revisit: When specific drug candidates are being evaluated. Replace the generic drug substrate with an agent-specific PK model (multi-compartment, metabolism, prodrug activation). The module architecture (Module 7) is already structured as a self-contained drug response unit; swapping the internal PK model doesn't affect other modules.

A5. Oxygen is supplied from the domain boundary as a proxy for vasculature.

What we assume: O_2 enters the simulation from the domain edge at a fixed concentration and diffuses

inward. There are no blood vessels, no perfusion dynamics, no vascular remodeling, and no distinction between arterial and venous supply.

What's really true: PDAC tumors have vasculature — sparse, collapsed, and dysfunctional, but present. Vessels supply O_2 and drug non-uniformly throughout the tumor. Stroma compresses vessels, creating avascular zones. VEGF-driven angiogenesis attempts (usually failing) to revascularize hypoxic regions.

Why this is acceptable: The boundary-as-vasculature approximation is valid when the tumor is smaller than the diffusion distance of O_2 from the nearest vessel. For a simulation domain of 5–10mm with O_2 supply at the edge, the O_2 gradient will resemble what real peri-tumoral vasculature produces: higher O_2 at the periphery, lower in the interior, with ECM density modulating the gradient steepness. The critical behavior — hypoxia developing in the tumor interior as stroma matures — is preserved. The approximation breaks down for very large tumors or for modeling anti-angiogenic therapy, neither of which is in scope for v1.0.

Risk if wrong: Drug delivery from the boundary is spatially uniform, but real vasculature delivers drug heterogeneously (some regions are better perfused than others). This means v1.0 may underestimate drug access in well-perfused subregions and overestimate it in avascular ones. The spatial sanctuary (Anchor 10) may be more uniform in the simulation than in reality, where sanctuary zones correspond to avascular pockets rather than the geometric center.

When to revisit: Cycle 3 (Vasculature). Add endothelial cell actors with vessel formation/collapse dynamics. Replace boundary O_2/drug supply with discrete vascular source points. The module architecture already handles spatially non-uniform substrate sources; the change is to the boundary condition specification, not to the PDE framework.

A6. EMT is a binary switch, not a continuous spectrum.

What we assume: A tumor cell is either epithelial (ZEB1=OFF, CDH1=ON) or mesenchymal (ZEB1=ON, CDH1=OFF). There are no intermediate or "partial EMT" states. The transition is gated by a single threshold.

What's really true: EMT is increasingly understood as a spectrum with stable intermediate states ("partial EMT," "hybrid E/M"). Cells can co-express epithelial and mesenchymal markers simultaneously. These hybrid states may have unique properties (collective migration, enhanced metastatic potential) not captured by a binary model.

Why this is acceptable: The stroma barrier problem requires EMT to produce two things: (a) spatial heterogeneity (mesenchymal front, epithelial core) and (b) MMP2-mediated local ECM degradation at the invasive front. Both are adequately captured by a binary switch with a signal-dependent threshold. The spatial pattern emerges because the threshold creates a sharp-enough boundary between EMT-positive and EMT-negative zones. Partial EMT states are most relevant for metastatic seeding and collective invasion — important for the biology of PDAC lethality, but not for the drug-access-through-stroma question.

Risk if wrong: If partial EMT cells have dramatically different secretion profiles (e.g., they secrete more TGF-β than either fully epithelial or fully mesenchymal cells), the simulation will miss a contributor to stroma buildup at the invasive front. There is some evidence for this, but the magnitude is uncertain.

When to revisit: If Reality Check 1 (natural history) shows unrealistic EMT spatial distributions — either too sharp a boundary or too diffuse — consider adding a PARTIAL EMT state with intermediate motility, intermediate adhesion, and possibly elevated TGF-β secretion. The `emt_extent` variable already supports categorical tiers (LOW/MED/HIGH); adding a PARTIAL tier is a parameter addition, not a structural change.

A7. TGF-β and SHH are the only stromagenic signals.

What we assume: Stroma activation is driven exclusively by TGF-β and SHH secreted by tumor cells (and TGF-β fed back by CAFs). No other cytokines, growth factors, or signaling axes activate stellate cells.

What's really true: PDAC tumor cells also secrete PDGF, FGF2, CTGF, IL-6, IL-1β, galectin-1, and other factors that contribute to stellate cell activation and ECM production. The stromagenic milieu is a complex cocktail, not a two-component signal.

Why this is acceptable: TGF-β and SHH are the two best-characterized and most pharmacologically targetable stromagenic pathways in PDAC. Together, they account for the majority of the desmoplastic response — TGF-β as the primary activation signal and SHH as the amplifying arm. Other factors (PDGF, FGF2, etc.) are additive contributors that reinforce the same downstream behavior (CAF activation, ECM production). Including them would increase model complexity (more PDE fields, more parameters, more uncertain kinetics) without changing the qualitative behavior of barrier formation. The two-signal model captures the essential dynamics: TGF-β builds the wall, SHH thickens it, and blocking either alone is insufficient.

Risk if wrong: If other stromagenic signals are so potent that they can sustain full barrier formation even when TGF-β AND SHH are both eliminated, the simulation will overestimate the efficacy of combined TGF-β/SHH inhibition. The EA might find dual-inhibition strategies that look excellent in simulation but fail in vivo because PDGF or IL-6 compensates. This is a real risk — stromagenic redundancy is one possible explanation for the failure of anti-stromal monotherapies in clinical trials.

When to revisit: If the EA consistently discovers strategies that eliminate the barrier by blocking TGF-β and SHH simultaneously, and these strategies seem "too good to be true," add a third stromagenic signal (PDGF is the strongest candidate) to represent pathway redundancy. If the EA's best strategies still work with the third signal present, they're robust. If they don't, the two-signal model was too optimistic.

A8. Mechanical forces are simplified to scalar pressure, not tensor stress.

What we assume: Solid stress from tumor growth is represented as a scalar pressure value at each location. There is no directional stress tensor, no shear stress, no viscoelastic relaxation, and no distinction between radial and circumferential stress components.

What's really true: Solid stress in tumors is a tensor quantity with directional components. Tumor growth generates compressive stress in the radial direction and tensile stress in the circumferential direction. Residual stress exists even after tumor excision. The mechanical environment is anisotropic, time-dependent, and coupled to ECM remodeling.

Why this is acceptable: The stroma barrier problem requires mechanical stress to do two things: (a) provide contact inhibition that restrains tumor growth (the cage function) and (b) compress ECM to

increase density passively (compaction). Both effects depend on the *magnitude* of stress, not its *direction*. A scalar pressure is sufficient to trigger contact inhibition when it exceeds a threshold and to drive ECM densification proportional to compression. Directional stress matters for predicting where the tumor will preferentially expand (path of least resistance) and how vessels are deformed — both important, but second-order for the diffusion-blocking question.

Risk if wrong: The tumor may expand isotropically in the simulation when it should preferentially invade along low-stress directions (e.g., along ducts, toward softer tissue planes). This could affect the spatial distribution of the invasive front and therefore the spatial pattern of drug exposure at the tumor-stroma boundary.

When to revisit: If the spatial pattern of tumor expansion in Check 1 (natural history) is unrealistically symmetric, consider upgrading to a 2D stress representation. PhysiCell's mechanics engine computes inter-cell forces as vectors; the upgrade path is to accumulate these vectors rather than collapsing them to a scalar.

A9. Time is discrete, not event-driven.

What we assume: The simulation advances in fixed time steps. All cells are evaluated every step. There is no event-driven scheduling where cells act only when something happens to them.

What's really true: Biological processes operate on vastly different timescales: signal diffusion (seconds), cell signaling (minutes), gene expression changes (hours), cell division (tens of hours), ECM accumulation (days). A fixed time step forces all processes to be evaluated at the same cadence, which is either too fast for slow processes (wasted computation) or too slow for fast processes (missed dynamics).

Why this is acceptable: PhysiCell uses an adaptive mechanics time step and a separate diffusion time step internally, which partially addresses this concern. The biological logic (module evaluations) runs at the mechanics step cadence, which is fine for the processes we model — the fastest biologically meaningful event is cell signaling response (~minutes), and PhysiCell's default time step (6 minutes for mechanics, smaller for diffusion) captures this. The stroma barrier evolves on a timescale of days to weeks, so we're not missing fast transients.

Risk if wrong: If drug pulses are very short (minutes), the time step resolution may not capture rapid PK dynamics. The EA might discover strategies with temporal precision that exceeds the simulation's resolution.

When to revisit: If the EA converges on very short pulse durations, decrease the mechanics time step or implement sub-step drug PK updates.

Part 2: Exclusions (Omitted Biology)

These are aspects of PDAC biology that do not exist in v1.0 at all — not simplified, just absent.

X1. Immune system.

What's excluded: All immune cell types — CD8+ T cells, CD4+ helpers, regulatory T cells, macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), natural killer cells, dendritic cells. No immune recognition, no immune killing, no immune suppression, no immune-related adverse events.

Why it's excluded: The immune system is the subject of Cycle 2. Including it in v1.0 would double the number of actors, modules, and edges before the stroma barrier physics are validated. The immune system interacts with the barrier (immune exclusion) but does not create it. The barrier forms through tumor-stroma paracrine signaling and ECM production — processes that are immune-independent. Validating the barrier first, then adding immune actors that interact with a known-correct barrier, is safer than debugging everything simultaneously.

Why it doesn't break the stroma barrier question: The barrier's ability to block drug delivery (the central question of Cycle 1) is a physical property of ECM density and diffusion. It doesn't depend on immune cells. Drug diffusion through stroma is the same whether T cells are present or not. The EA's task — find interventions that breach the barrier and kill tumor cells with drug — is fully defined without immune actors.

What's preserved for Cycle 2: Immune-evasion tendency (IET) is computed and tracked per-cell via ZEB1 and HIF1A states. PD-L1 status is bookkeeping. When immune actors arrive, they read these pre-

computed values. No Cycle 1 architecture needs to change; Cycle 2 adds actors and one new module.

When it enters: Cycle 2, explicitly.

X2. Vasculature and angiogenesis.

What's excluded: Blood vessels as discrete structures, endothelial cell actors, VEGF-driven angiogenesis, vessel co-option, vascular collapse dynamics, perfusion heterogeneity, and anti-angiogenic therapy.

Why it's excluded: Vasculature adds at least two new actor types (endothelial cells, pericytes), one new PDE field (VEGF), and a complex morphogenesis model for vessel formation and pruning. This is a simulation-within-a-simulation and would dominate the debugging effort. The boundary-as-vasculature approximation (Assumption A5) is adequate for the stroma barrier question because drug delivery limitation in PDAC is driven by ECM-mediated diffusion blockade more than by vascular insufficiency — the drug reaches the peri-tumoral vessels, it just can't get through the stroma after that.

Why it doesn't break the stroma barrier question: The barrier blocks drug *after* it exits the vasculature. Whether the vasculature delivers drug uniformly (our approximation) or heterogeneously (reality), the ECM still determines how far the drug penetrates beyond the vascular source. The essential competition — drug diffusion rate vs. ECM density — is preserved.

When it enters: Cycle 3 or when the EA's discovered strategies require precise drug delivery modeling (e.g., strategies that exploit vascular heterogeneity or anti-angiogenic priming).

X3. Metabolism and nutrient competition.

What's excluded: Glucose uptake, lactate production, the Warburg effect, amino acid deprivation, lipid metabolism, autophagy as an explicit survival mechanism, and all metabolic reprogramming pathways except HIF1A-gated stress response.

Why it's excluded: Full metabolic modeling would require additional PDE fields (glucose, lactate, glutamine at minimum), metabolic flux calculations per cell, and extensive parameterization from

metabolomics data that is sparsely available at single-cell resolution in PDAC. The computational cost of adding 3+ PDE fields is substantial. The metabolic consequences that matter for the stroma barrier — hypoxia-driven adaptation, stress resistance — are adequately captured by the O_2 field and HIF1A response (Assumption A5, Module 1).

Why it doesn't break the stroma barrier question: The barrier blocks drug delivery via diffusion impedance. Metabolic competition (e.g., CAFs and tumor cells competing for glucose) affects cell viability over long timescales but does not change the physical properties of ECM or the diffusion physics that determine drug access. A metabolically detailed model would produce more realistic cell death patterns but would not change the fundamental barrier-formation or drug-exclusion dynamics.

When it enters: Cycle 3 (Metabolic Resistance), specifically when the EA's strategies need to be validated against metabolic adaptation (e.g., if the EA discovers strategies that rely on starving tumor cells by cutting off nutrient supply through stroma modulation).

X4. Metastasis and distant spread.

What's excluded: Intravasation, circulation, extravasation, colonization of distant organs, and the entire metastatic cascade. The simulation models a single primary tumor site. Cells that reach the domain boundary do not "metastasize."

Why it's excluded: Metastasis is how PDAC kills most patients, but it is not the stroma barrier problem. The stroma barrier is a local phenomenon at the primary tumor site — it determines whether locally delivered drug can reach locally resident tumor cells. Metastasis involves entirely different biology (dissemination, dormancy, colonization) that operates on different spatial and temporal scales.

Why it doesn't break the stroma barrier question: Drug penetration through peri-tumoral stroma is a local physics problem. It does not depend on whether distant metastases exist. The EA's task is to find ways to breach the barrier at the primary tumor. Whether those strategies also affect metastatic sites is a separate clinical question that v1.0 does not address.

When it enters: Not currently planned for any cycle. Metastasis modeling is a separate project with its own simulation requirements, not an extension of the stroma barrier world.

X5. Neural invasion and perineural spread.

What's excluded: Nerve fibers in the tumor microenvironment, Schwann cells, neurotrophin signaling (NGF, GDNF), and the perineural invasion phenotype.

Why it's excluded: Perineural invasion is a hallmark of PDAC and a major contributor to pain and local recurrence. However, it is a migration/invasion problem, not a barrier problem. Neural invasion occurs along pre-existing nerve tracts that provide a low-resistance path for tumor cells — the opposite of the stroma barrier, which is a high-resistance obstacle. Including neural structures would require new actor types and signaling pathways that do not interact with the barrier-formation or drug-exclusion questions.

Why it doesn't break the stroma barrier question: Nerves do not produce ECM, do not modify drug diffusion, and do not participate in the TGF-β/SHH paracrine loop. Their presence or absence does not change how drug penetrates stroma.

When it enters: Not currently planned. If invasion dynamics become a focus in a future cycle, neural structures could be added as directional migration cues in the ECM field.

X6. Exosomes and extracellular vesicles.

What's excluded: Tumor-derived exosomes, their cargo (miRNA, proteins), and their roles in pre-metastatic niche formation, immune modulation, and stromal education.

Why it's excluded: Exosome signaling operates on systemic scales (preparing distant organs for metastasis) and at molecular scales (miRNA-mediated gene regulation) that are both outside v1.0's modeling resolution. Local paracrine signaling via diffusible TGF-β and SHH captures the dominant tumor→stroma communication mechanism.

Why it doesn't break the stroma barrier question: The stromal activation that builds the barrier is driven by diffusible cytokines, not by exosomal cargo. Exosomes may fine-tune stromal behavior at the level of CAF subtype specification, but the gross barrier-formation dynamics (PSC activation, ECM production) are cytokine-driven.

When it enters: Unlikely to enter any near-term cycle unless evidence emerges that exosomal signaling is required for barrier maintenance (as opposed to modulation).

X7. Lymphatic system and interstitial fluid dynamics.

What's excluded: Lymphatic vessels, interstitial fluid pressure (IFP) as a dynamic variable, convective drug transport, and lymphatic drainage.

Why it's excluded: IFP in PDAC is extremely high (driven by HA-rich stroma collapsing lymphatics and vasculature). This is one of the major mechanisms of drug exclusion. However, v1.0 approximates this effect through ECM density → diffusion coefficient coupling (specifically, the HA fraction reduces diffusion rate). The physical effect of high IFP — opposing convective drug transport — is qualitatively captured by making drug diffusion slower in dense ECM. Explicit IFP modeling would require a fluid dynamics simulation coupled to the tissue mechanics, which is computationally expensive and parameterically uncertain.

Why it doesn't break the stroma barrier question: The barrier's drug-blocking effect is already modeled through diffusion impedance. The approximation converts a pressure-driven (convective) blockade into a diffusion-rate-driven blockade, which produces the same qualitative outcome: drug reaches the tumor more slowly when the stroma is dense. The quantitative details differ (convection vs. diffusion have different spatial profiles), but the Reality Anchors are all qualitative tests (drug penetration decreases with barrier maturity, monotonically). They will pass with either mechanism.

When to revisit: If quantitative drug delivery predictions become important (e.g., predicting absolute drug concentrations rather than relative gradients), IFP modeling would be necessary. Consider a Darcy flow model coupled to the ECM density field.

X8. Epigenetic regulation and non-coding RNA.

What's excluded: DNA methylation dynamics, histone modification, chromatin remodeling, miRNA regulation of gene expression, lncRNA-mediated pathway modulation.

Why it's excluded: The gene state representation (WT/GOF/LOF) abstracts over the mechanism of gene alteration. A gene silenced by promoter methylation and a gene deleted by homozygous loss are both LOF in our system. The downstream phenotypic effect is the same for the purposes of the Boolean network. Modeling the mechanism of inactivation adds complexity without changing the phenotype-level behavior that drives barrier formation.

Why it doesn't break the stroma barrier question: The stroma barrier is built by cells acting on their current phenotype, which is determined by gene states + microenvironmental signals. Whether a gene reached its current state through mutation, deletion, or epigenetic silencing does not change what the cell does at a given time step.

When to revisit: If therapy-induced epigenetic changes (e.g., demethylation agents reactivating silenced tumor suppressors) become part of the EA's intervention space, the gene state model would need to support state transitions (LOF → WT via epigenetic reactivation). This is a natural extension of Rule 12 (deferred clonal evolution).

Part 3: Risk Ranking

Not all assumptions and exclusions carry equal risk. Here they are ranked by the likelihood that they will produce misleading results in v1.0:

Rank	ID	Item	Risk level	Reason
1	A7	Only TGF-β and SHH as stromagenic signals	**HIGH**	If stromagenic redundancy is strong, the EA will overestimate the efficacy of pathway-targeted stroma interventions. This is the assumption most likely to produce clinically misleading results.
2	A1	Monoclonal tumor	**MEDIUM**	Matters if resistance depends on pre-existing subclones rather than inducible adaptation. v1.0's NRF2/ABCB1 mechanism captures inducible resistance but not genetic heterogeneity.
3	X1	No immune system	**MEDIUM**	By design (Cycle 2), but means v1.0 cannot evaluate strategies that depend on immune synergy (e.g., stroma depletion + checkpoint inhibitor). The EA's discoveries will be drug-only.
4	A5	Boundary-as-vasculature	**MEDIUM**	Drug delivery is spatially uniform in v1.0. Real vasculature creates pockets of good and bad perfusion. Sanctuary zones in v1.0 are geometrically simple; real sanctuaries are shaped by vascular anatomy.
5	A4	Generic drug PK	**LOW-MED**	Matters for clinical translation (timing, dosing) but not for discovering barrier-breaching principles.
6	A2	Irreversible CAF activation	**LOW**	Conservative error (barrier is harder to eliminate in simulation than in reality).
7	X7	No interstitial fluid pressure	**LOW**	Qualitatively captured by ECM→diffusion coupling. Quantitative error only.
8	A3	Scalar ECM density	**LOW**	Sufficient for diffusion blocking. Fiber orientation matters for directional migration, which is secondary.

Rank	ID	Item	Risk level	Reason
9	A6	Binary EMT	**LOW**	Captures the spatial pattern. Partial EMT states are refinements, not foundational.
10	A8	Scalar stress	**LOW**	Magnitude-dependent contact inhibition is the required behavior. Directionality is secondary.
11	X2–X6, X8	Vasculature, metabolism, metastasis, nerves, exosomes, epigenetics	**LOW**	These exclusions are far from the stroma barrier mechanism. They would matter for a comprehensive PDAC model but not for the specific question of drug diffusion through stroma.

Part 4: The "Why Not Just Include Everything" Argument

The temptation is to add more biology because more biology is more realistic. Here is why that impulse is wrong for v1.0:

Every additional PDE field costs real computation. Going from 4 diffusible substrates to 8 roughly doubles the PDE solver time, which is already the bottleneck. The EA needs thousands of simulation runs. Doubling the cost per run halves the EA's search budget or doubles the LARCC allocation.

Every additional parameter needs calibration data. The 13 knobs in the current spec can be anchored to published cell line and patient data. Adding PDGF signaling, glucose metabolism, and lymphatic dynamics would add 10–20 more knobs, most of which have sparse or inconsistent calibration data in PDAC. Poorly calibrated parameters produce confident but wrong results — the most dangerous kind of error.

Every additional module interaction creates new failure modes. The current 8-module system has 28 directed edges and 5 feedback loops. Adding 3 more modules could easily double the edge count and introduce feedback loops that produce oscillatory, chaotic, or numerically unstable behavior. Each new

loop needs its own set of Reality Anchors to validate. The debugging time scales super-linearly with system complexity.

The stroma barrier question has a finite scope. We are not building a general PDAC simulator. We are building a world where the EA can discover how to breach a physical barrier and kill the cells behind it. The barrier is made of ECM produced by CAFs activated by TGF-β and SHH. The drug is blocked by the ECM. The tumor resists with efflux. The barrier rebuilds from surviving CAFs. That is the complete problem statement. Everything else is a different problem for a different cycle.

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30 items: 9 assumptions, 8 exclusions, 1 risk ranking, 1 philosophy statement. Every simplification is a debt. This section is the ledger. Pay it down in Cycles 2–3, not before the stroma barrier is solved.